

- 3 (a) (work =) force $\times$ distance moved / displacement in the direction of the force
OR when a force moves in the direction of the force work is done B1 [1]
- (b) kinetic energy = $\frac{1}{2} mv^2$
= $\frac{1}{2} 0.4 (2.5)^2 = 1.25 / 1.3 \text{ J}$ C1
A1 [2]
- (c) (i) area under graph is work done / work done = $\frac{1}{2} Fx$ C1
 $1.25 = (14 x) / 2$ C1
 $x = 0.18 (0.179) \text{ m}$ [allow $x = 0.19 \text{ m}$ using kinetic energy = 1.3 J] A1 [3]
- (ii) smooth curve from $v = 2.5$ at $x = 0$ to $v = 0$ at Q M1
curve with increasing gradient A1 [2]

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